The gain will be more for B than for A. On the contrary, if B's demand for X commodity is less elastic, the terms of trade will be closer to the domestic exchange ratio of country B: 1 unit of X = 1.5 unit of Y. The terms of trade, in this situation, will be favourable for A and against B. Country A will have a larger share out of the gains from trade than country B.

The distribution of gains from trade can be explained in terms of the Marshall-Edgeworth offer curve through Fig. 2.3.

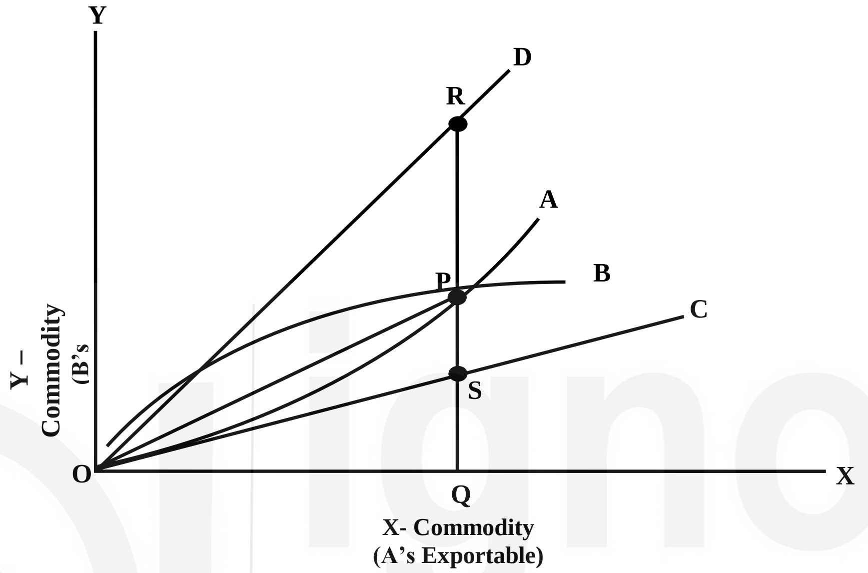

Figure 2.3: Marshall-Edgeworth offer curve

In Fig. 2.3., OC and OD are the domestic exchange ratio lines of countries A and B respectively. OA is the offer curve of country A and OB is the offer curve of country B. The exchange takes place at P where the two offer curves cut each other. Country A imports PQ quantity of Y and exports OQ quantity of X.

The terms of trade for country A at P = (Q_Y/Q_X) = (PQ/OQ) = Slope of Line OP. If the line OP gets closer to OD, the terms of trade become favourable to country A and unfavourable to country B. On the opposite, if the line OP gets closer to the line OC, the domestic exchange ratio line of country A, the terms of trade turn against country A and become favourable to country B.

Country A was willing to exchange before trade SQ units of Y for OQ units of X. After the trade, it gets PQ units of Y for OQ units of X. Therefore, the gain from trade for country A, out of the total trade gain of RS, amounts to PQ – SQ = PS units of Y. In the case of country B, RQ units of Y were being exchanged for OQ units of X before the trade.

However, after trade it has to part with only PQ units of Y to import OQ units of X. Therefore, the gain from trade for this country amounts to RQ – PQ = RP units of Y. As the point of exchange, P gets closer to the line OD, the share of country A in the gain from trade will rise and that of country B will fall and vice-versa.

4) **Taussig's Approach:**

According to Taussig, trading country gains from trade in form of a rise in income. As the demand for labour increases because of increase in exports, employers offer higher wages to attract more labour in this industry. The spill over effect of rise in wages in one industry is that other industries will also have to offer higher wages to retain talent. Thus international trade, according to Taussig will lead to a general rise in money incomes. A higher level of income due to trade enables the people of a country to make larger purchases of both domestically produced and imported goods and reach a higher level of welfare.

5) **Modern Approach:**

The modern theorists considered the gains from trade as the gains resulting from:

i) Gains from exchange and

ii) Gains from specialisation.

These two gains together constitute the gains from international trade. When trade take place between countries, consumers enjoy a higher level of satisfaction, partly because of improvement in terms of trade and partly on account of greater specialisation in the use of economic resources of the country. These two types of gains from trade can be shown in Fig. 2.4.

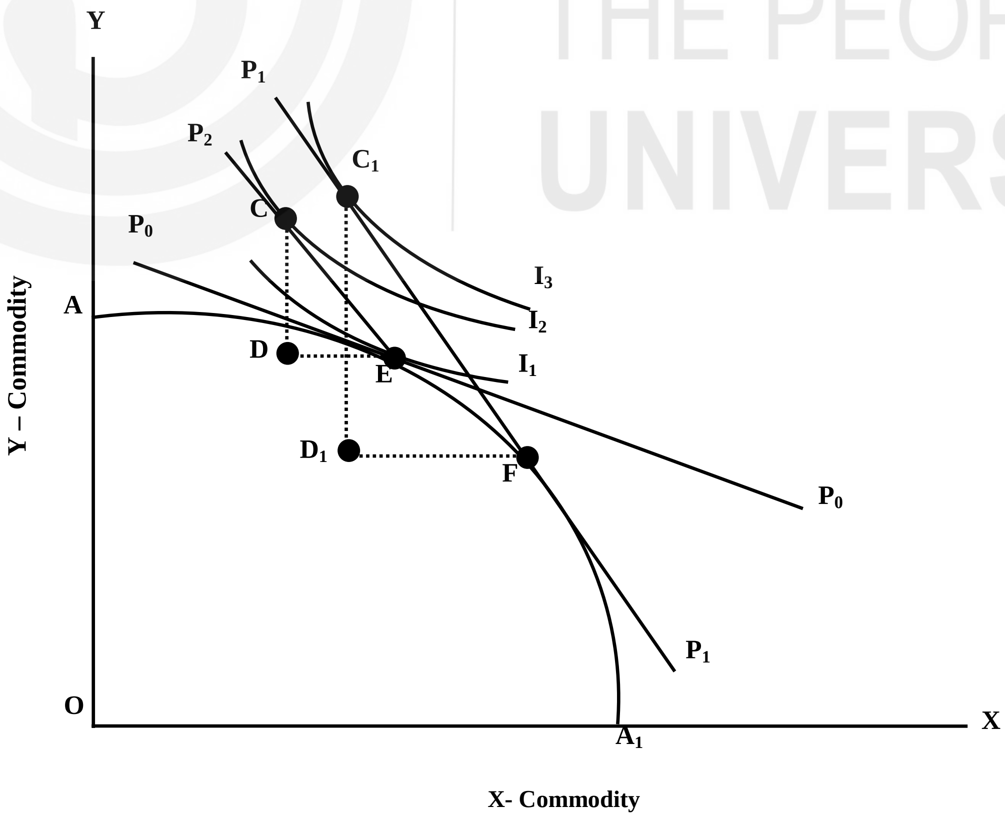

Figure 2.4: Gains from trade

In Fig. 2.4., X-commodity is measured along the horizontal scale and Y-commodity is measured along the vertical scale. AA_1 is the production possibility curve. P_0P_0 is the domestic price ratio line. It is tangent to the production possibility curve at E. Thus, E is the point of production equilibrium in the absence of trade.

E is also the point of consumption equilibrium because P_0P_0 is tangent to the community indifference curve I_1 at this point. When trade commences, P_1P_1 is the international exchange ratio line, which is tangent to the production possibility curve at F and the community indifference curve I_3 at C_1.

Thus, F is the point of production and C_1 is the point of consumption. After trade takes place, D_1F of X-commodity is exported and C_1D_1 quantity of Y-commodity is imported. The trade causes two types of shifts in the country. First, the production point shifts from E to F. It occurs because of specialisation in the production of X-commodity and specialisation in factor use. This can be called the production effect. Second, the point of consumption shifts from E at I_1 to C_1 at the higher community indifference curve I_3. It means an increase in the satisfaction of the commodity. This can be called the consumption effect. Both together signify the gain from international trade.

If a line P_2E is drawn parallel to P_1P_1 from the original equilibrium situation E, it signifies that there is no change in production, but the consumption equilibrium shifts from E to C at a higher community indifference curve I_2. In this situation, the CD quantity of Y is imported at a lower international price of Y. The quantity of X-commodity exported in exchange for CD quantity of Y is DE. Although production is the same as at point E, the consumption equilibrium shifting from E to C signifies the gain from trade. This is the trade gain from the exchange.

After the trade, as the specialisation in production and optimum factor use takes place, the production equilibrium shifts from E to F along the same production possibility curve and the consumption equilibrium shifts to C_1. In this situation, C_1D_1 quantity of Y is imported and D_1F quantity of X is exported. As a result of specialisation in production after the trade, the shift in consumption equilibrium from C to C_1 reflects the trade gain from specialisation.

To sum up, the total gain from trade is comprised of gain from the exchange and the gain from specialization. The total gain from trade can be measured by the movement from E to C_1. This movement takes place in two steps—the movement from E to C is the gain from the exchange and the movement from C to C_1 is the gain from specialization.

Another Example:

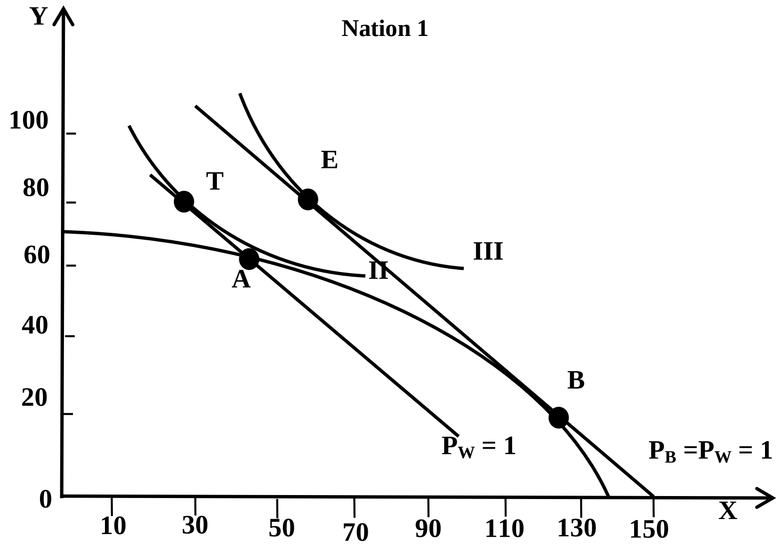

Figure 2.5: Gains from specialisation in production

If Nation 1 could not specialize in the production of X with the opening of trade but continued to produce at point A, Nation 1 could export 20X in exchange for 20Y at the prevailing world price of $P_W = 1$ and end up consuming at point T on indifference curve II. The increase in consumption from point A (in autarky) to point T represents the gains from exchange alone. If Nation 1 subsequently did specialize in the production of X and produced at point B, it would then consume at point E on indifference curve III. The increase in consumption from T to E would represent the gains from specialization in production.

6) **Gains from Trade for Large Country versus Small Country:**

H.R. Heller explained how entire gains go to a small country instead of large country under the conditions of constant opportunity cost and unchanged terms of trade. This case can be explained in Fig. 2.6.

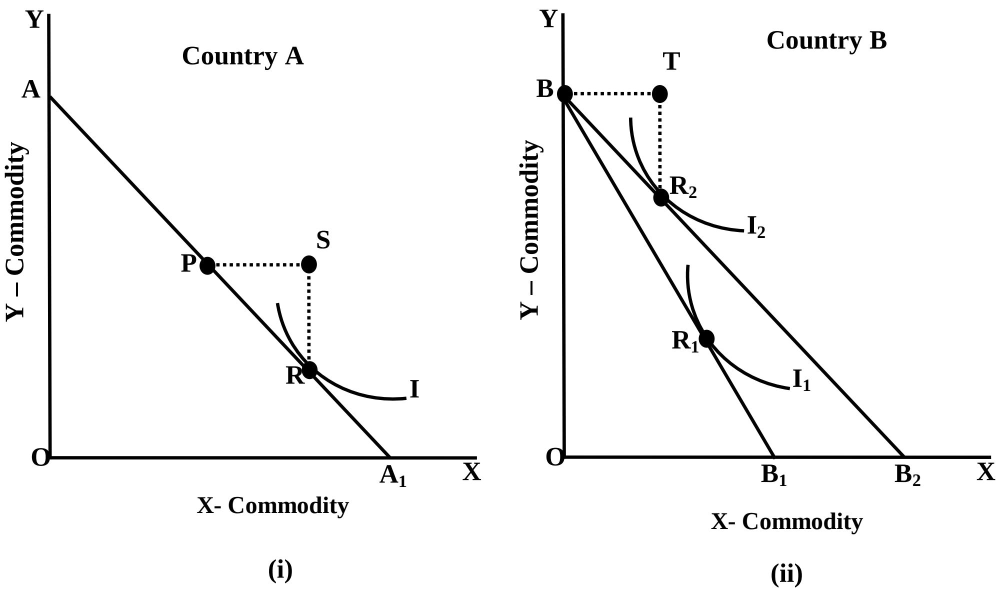

Fig. 2.6. (i) & 2.6. (ii): Gains from Trade for Large Country and Small Country

In Fig. 2.6. (i) For a large country A, the production possibility curve under the conditions of constant costs is AA_1. In the absence of trade, consumption and production take place at R where the community indifference curve I is tangent to the production possibility curve. After trade takes place, there is no change in terms of trade for country A so that the international price ratio line remains AA_1. This country will, however, modify its production pattern in such a way that some imports are made from country B. It may decide to move to P where it exports PS quantity of X commodity and imports SR quantity of Y. Since the terms of trade remain unchanged for country A, it fails to make any gain from trade.

In Fig. 2.6. (ii), for the small country B, the production possibility curve or domestic price ratio line under constant cost conditions is BB_1. Its tangency with the community indifference curve I_1 shows that production and consumption equilibrium in this country, in the absence of trade, takes place at R_1. As trade commences, this country specialises completely in the production of Y commodity. The international price ratio line is BB_2, which is parallel to AA_1. This country produces at B. The consumption equilibrium occurs at R_1.

So, after trade it exports TR_2 (= SR) of Y commodity to country A and imports BT (= PS) quantity of X from country A. The movement from R_1 to R_2 in country B reflects the gain from specialisation and exchange to the small country B from the international trade. Since this country can import X-commodity at a lower international price, the terms of trade turn in favour of it. That also shows that the gains from trade go to small country B alone and the large country goes without any gain from trade.

2.7 POTENTIAL AND ACTUAL GAIN FROM TRADE

Domestic cost ratios of producing two commodities X and Y determines the potential gain from trade for the two trading countries A and B.

Thus,

$$G_P = \left(\frac{C_X}{C_Y}\right)_A - \left(\frac{C_X}{C_Y}\right)_B$$

Where G_P = potential gain C_X = cost per unit (money or labour) for X commodity, C_Y = Cost per unit (money or labour) for Y commodity. The subscripts 'A' and 'B' signify the two countries.

The actual gain from trade G_a, on the other hand, is determined by the difference in price ratio of the two commodities X and Y in the two countries.

Thus,

$$G_a = \left(\frac{P_X}{P_Y}\right)_A - \left(\frac{P_X}{P_Y}\right)_B$$

Under the conditions of perfect competition and free trade, the cost ratio equals the price ration of two commodities in each country such that

$$\frac{C_X}{C_Y} = \frac{P_X}{P_Y}$$

Consequently the actual gain becomes equal to potential gain

$G_P = G_a$

In absence of tariffs and other trade restrictions, actual gains are equal to potential gains of the country. In case of presence of imperfect competition or trade and tariff restrictions, the price ratio (P_X/P_Y) is more than the cost ratio (C_X/C_Y) of a country. Thus, the actual gain from trade exceeds the potential trade gain ($G_a > G_P$).

2.8 FREE TRADE VERSUS NO TRADE

When there are no tariffs or trade restrictions, it is called a situation of free trade. In situation of free trade, domestic factor and product prices get equalised with international prices.

Assumptions:

The proposition that free trade is superior to no trade is proved based on the following assumptions:

a) There is a situation of perfect competition in the market.

b) The government does not interfere in trade i.e. there are no tariffs, quotas and subsidies.

c) There are no transportation costs.

d) The given country is small in size and has no monopoly power in trade.

e) The factors of production are fixed in supply.

f) The technology is such that the production possibility curve is concave to the origin.

Based on the assumptions given above, it is possible to show that free international trade is much superior to autarchy (absence of trade). The diagram, to demonstrate it, is adapted from the diagram given by Jagdish Bhagwati.

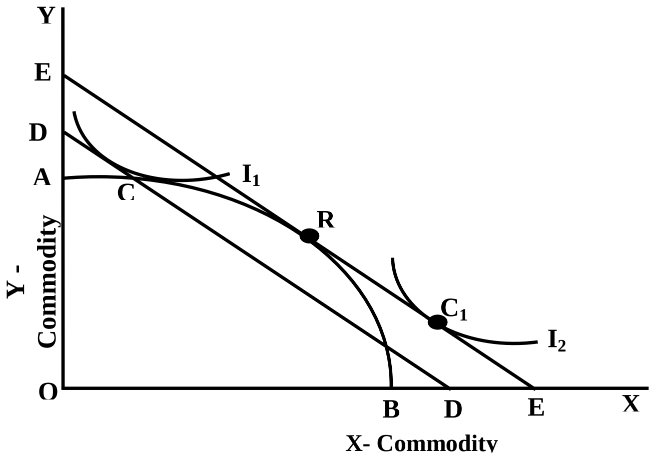

Figure 2.7: Benefits of free trade vs. no trade

In Fig. 2.7 AB is the production possibility curve of the home country. In the absence of trade, the domestic price ratio is given by the line DD. C is the point of production and consumption equilibrium. As the trade commences and there is no restriction on trade, the international price ratio is given by the slope of the line EE which runs parallel to DD. The line EE represents the Consumption Possibility Curve.

EE determines the new availability frontier in the country. Point R, where the consumption possibility curve is tangent to the production possibility curve, represents the most efficient production point. Consumption point on the other hand is determined at C_1 where the international price ratio line EE is tangent to the higher community indifference curve I_2. Since after free trade, the production is optimised at R and consumption is optimised at C_1, it follows that free trade is superior to no trade.

2.9 STATIC AND DYNAMIC GAINS

The gains from international trade are of two types:

1) **Static Gains & Dynamics Gains**

   The static gains from trade are as under:

   i) **Division of labour, specialization and Expansion in Production**: International trade accord the benefit of division of labour and specialization. It ensures that all the available productive resources in the country are optimally utilized leading to expansion in production of the firm, country and the world.

   ii) **Improved Welfare**: Increased production because of international trade leads to improved welfare in the trading countries. Quality of the products also improves due to specialisation by a country. Quality along with easy access at a lower costs directly boosts welfare of the people.

As Ricardo points out "The extension of international trade very powerfully contributes to increasing the mass commodities and, therefore, the sum of enjoyments."

iii) **Increase in National Income**: International trade leading to expansion of production in the trading countries helps to generate more and more employment opportunities. Increase in production and employment leads to a rise in the national income of the trading countries.

iv) **Vent for Surplus**: According to Adam Smith, international trade leads to the fullest utilisation of productive resources of the country. The surplus created because of increased efficiency and increased production can be utilized through sales in the foreign market.

2) **Dynamic Gains from Trade:**

The major dynamic gains from international trade are as follows:

i) **Technological Development**: International trade facilitates transfer of advanced technology from developed to less developed countries. It stimulates innovations and inventions which in turn bring efficiency in production and reduces overall cost. Latest and advanced techniques are transferred to other countries through trade.

ii) **Increased Competition**: Along with other benefits, international trade brings in efficiencies in all the industries in trading countries. It creates the environment of competition ultimately leading to increased quality of goods.

iii) **Expansion of Market**: International trade leads to expansion in the size of the market. It encourages producers to expand the scale of production which in turn leads to the increased volume of investment and employment. Increased investment and employment ultimately lead to expansion in the market.

iii) **Increase in Investment**: As the demand for home-produced goods increases due to international trade, there is a strong impetus for investment. The growth of the export sector leads to the expansion of several allied ancillary industries creating more and more opportunities for investment. There is also a substantial increase in foreign direct investments in the export sector of the economy.

v) **Optimum Use of Resources**: International trade leads to more efficient use of all the productive resources.

Ellsworth and Clark Leith summed up the dynamic gain from trade in these words, "Trade is a dynamic force that stimulates innovation. New ways of producing and organising production are spread to the local economy through trade and the competitive force of trade stimulates the adoption of cost-saving techniques. Trade also makes possible economical local production of many goods that would be prohibitive to produce locally."

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) In what circumstances might a country NOT benefit from trade with another country?

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2) What are static and dynamic gains from trade?

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3) Explain Mill's approach to measuring gains from trade.

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4) Why is ‘free trade’ between two countries is considered to be better than ‘no trade’?

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2.10 LET US SUM UP

After reading this unit, we learnt :

a) To maximize the value of its output, a country must be producing a combination of goods and services that lies on its production possibilities curve.

b) Suppose two countries each produce two goods and their opportunity costs differ. If this is the case, there is an opportunity for trade between the two countries that will leave both better off.

c) International trade leads countries to specialize in goods and services in which they have a comparative advantage.

d) The terms of trade determine the extent to which each country will specialize. Each will increase the production of the good or service in which it has a comparative advantage up to the point where the opportunity cost of producing it equals the terms of trade.

e) Free international trade can increase the availability of various goods and services in the countries that participate in it. Trade between the countries allow consuming combinations of goods and services which they would not be able to produce.

f) While free trade increases the availability of total quantity of goods and services to each country, in short run, there might be winners and losers.

2.11 KEYWORDS

Autarky : The absence of trade; economic isolation

Absolute Advantage : The greater efficiency that one nation may have over another in the production of a commodity. It describes a situation in which a country can produce more of a good or service than any other with the same quantity of resources. According to Adam Smith, this is the basis for trade.

The basis for Trade : The forces (absolute advantage for Adam Smith and Comparative Advantage for Ricardo).

Specialisation : When a country allocates most or all its resources towards the production of a particular good or service. For example, Bangladesh specializes in producing textiles.

Gains from Trade : The increase in each nation's consumption resulting from specialization in production and trade.

Terms of Trade : The price of one good in terms of the other that two countries agree to trade at. Beneficial terms of trade allow a country to import a good at a lower opportunity cost than the cost for them to produce the good domestically, thus the country gains from trade. In other words, it's the ratio of the price of a nation's export commodity to the price of its import commodity. When more than two commodities are traded, we use the index of export to import prices. Terms of trade are usually given in percentages.

Consumption Frontier : The various alternative combinations of the two commodities that a nation can consume.

Marginal rate of Transformation (MRT)

The amount of one commodity that a nation must give up to get more units of a second commodity. This is another name of the opportunity cost of a commodity. To be more precise, marginal rate of transformation shows the number of goods which will be given up by a country to produce an additional unit of other goods while keeping the factors of production constant.

Production Possibilities Curve

It is a graph which shows various combinations of output that a country can produce by applying its best available resources and technology. In other words, A curve that shows the various alternative combinations of the two commodities that a nation can produce by completely utilising all its factors of production with the best available technology. This curve is also known as the transformation curve or the production frontier.

Opportunity Cost Theory

This opportunity cost theory states that the cost of one commodity is the amount of another commodity that must be given up by a country to release just enough factors of production or resources to enable the production of one additional unit of the first commodity. To be more precise, if a country can produce either commodity A or B, the opportunity cost of commodity A is the amount of the other commodity B that must be given up to get one additional unit of commodity A.

Constant Opportunity Costs

It occurs when the opportunity cost stays the same as a country increase its production of one good. This shows that the resources are easily accommodated or changed from the production of one good to the production of another good. To be more precise, constant opportunity costs theory occurs when the equal amount of one commodity that a nation give up releasing just enough resources to produce each additional unit of another commodity.

Increasing Opportunity Costs

It describes how opportunity costs for a country increase as resources are applied. According to the law of increasing opportunity costs, as one good is produced, the opportunity cost to